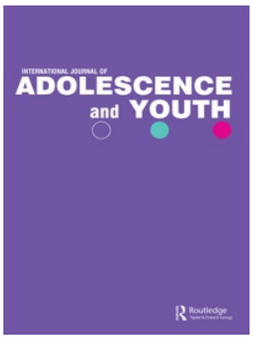

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REVIEW ARTICLE

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Short video addiction and its impact on cognitive functioning in adolescents and youth: a systematic review

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ABSTRACT

Short-video platforms are rapidly growing worldwide, raising concerns about their cognitive effects. While several reviews have explored short-video addiction (SVA) and mental health, research on cognitive outcomes remains limited. This review synthesises findings from 23 studies examining SVA and its cognitive consequences in adolescents and young adults. Attention and self-control were the most affected cognitive domains, with additional but less frequent impacts on memory, decision-making, negative cognitive bias, loss of aversion, and cognitive engagement. SVA was associated with attentional deficits, impaired executive control, and working memory problems, which may undermine academic performance and adaptive behaviour. Self-control emerged as a protective factor, whereas impulsivity, boredom, and negative cognitive bias increased vulnerability. The findings highlight the importance of interventions promoting self-control and healthy short-video use to mitigate cognitive impairments. In the present study, most included research was conducted in China, emphasising the need for studies in other countries to cover diverse populations.

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Short video addiction; cognitive functioning; attention; self-control; adolescents; young adults

Introduction

Social media has an all-pervasive presence in the lives of adolescents and young adults, across the world. It has both positive and adverse effects among young people. While social media helps young people connect, engage and interact with each other; build a sense of community and make meaning of the world around them (Cole et al., 2017; Marchi & Clark, 2021; Potdar et al., 2022); problematic social media use is also on the rise. Problematic social media use significantly affects life satisfaction, familial relationships, self-esteem, (Çiçek et al., 2024; Çiçek et al., 2025; Ünsal et al., 2025; Yıldırım et al., 2025) workplace focus (Wang, 2024), academic performance (Giunchiglia et al., 2017) as well as cognitive functioning (Dikshit & Kiran, 2023). One form of problematic social media use growing exponentially is short video addiction (SVA).

Short videos are defined as videos that are highly engaging for users, readily watchable, diverse in content form, and limited to a few minutes in duration (Feng, 2024). They are a thriving form of entertainment (Chen et al., 2022; Wang, 2020), that provide viewers with convenient access to news and information (Wang, 2021). Short videos keep the viewers engaged and intrigued by the content by employing theatrical tools such as situational construction, the cultivation of suspense, and the use of mimesis (Wang & Suthers, 2022).

These engagement features are inherent to the design of the short-form video format, which first emerged in 2013 with the launch of Vine, a video-sharing application where users recorded short videos of up to six seconds (Valley, 2023). This has grown into a phenomenon, driven by platforms such as TikTok, Instagram Reels, Snapchat's Snap Stories, and YouTube Shorts. The popularity of short-form content is rising

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astronomically. In 2025, the global TikTok user base is estimated to reach approximately 955.3 million users (Statista, 2025). Similarly, YouTube Shorts has achieved extensive reach, with over 2 billion monthly active users and an estimated 70 billion daily views (Kumar, 2025). Despite the widespread adoption of short-form video platforms, growing evidence suggests that their increased use is associated with heightened risk of addictive behaviours (Jung et al., 2025).

Short Video Addiction (SVA), also referred to as short form video addiction, problematic short video use, or excessive use of short-form videos, is characterised by an uncontrollable urge to watch, difficulty regulating usage, and compulsive viewing (Li et al. 2025; Liao, 2024). It has the potential to be addictive due to the type of content, the platform function hooks, personal interests, habitual viewing, and the need for social interaction and reassurance (Yang et al., 2021). Although these videos may offer momentary emotional relief, excessive engagement has been linked to increased rates of depression, fear of missing out, lower core self-evaluation, sleep disturbance, and self-injurious behaviours, behavioural dysregulation and impairments in all domains of life (Liao, 2024; Liu et al., 2024; Yu et al., 2024). The cognitive functioning of individuals, particularly adolescents and young adults, is a fundamental area in which SVA has a significant impact (Al-Leimon et al., 2025; Kim, 2024; Xu et al., 2023).

Cognitive functions refer to the internal mental processes that determine how individuals perceive, remember, speak, think, make decisions, and solve problems (Roy, 2020). These consist of executive function, social cognition, language, orientation, memory, attention, and consciousness (Miura & Hattori, 2014). Multiple studies provide converging evidence that cognitive control continues to undergo change between late adolescence and young adulthood (Friedman et al., 2016; Knežević, 2024; Luna et al., 2010; Veroude et al., 2013). Studies have revealed primarily a complicated and an intricate relationship between cognitive functioning and digital media use (Hietajärvi et al., 2022; Reid Chassiakos et al., 2016; Veroude et al., 2013). For most part, the nature of this relationship is adverse with excessive digital media use affecting cognitive control in teens and young adults (Alho and Moisala, 2022; Luna et al., 2010; Müller & Scherer, 2022; Onyeaka et al., 2022).

Review of literature

Recent research has highlighted the cognitive risks associated with digital media use, leading to growing interest in the effects of short-form video platforms. Empirical evidence shows that frequent exposure to brief, rapidly presented videos is linked to reduced attentional stability, impaired inhibitory control, and increased cognitive fragmentation (Chen et al., 2022; Jiang et al., 2024; Pan et al., 2025). SVA has also been associated with disrupted cognitive regulation, compulsive reward-driven behaviours, and difficulties in maintaining sustained mental engagement (Qin et al., 2022; Xie et al., 2023). Adolescents and young adults appear particularly vulnerable, as prolonged engagement with algorithmically curated video streams can lead to cognitive overload and attentional dysregulation (Eun, 2025; Hoffman, 2024; Nakhla, 2025; Wang, 2025a). These findings highlight the cognitive implications of SVA and the importance of examining its effects in contemporary digital media environments.

Research consistently shows that attention-related impairments are the most significant cognitive consequence, with individuals exhibiting greater deficits in attentional concentration and increased processing interference during short video consumption (Chen et al., 2022). Neurophysiological evidence, including alterations in the P300 component, indicates disruptions in attentional allocation and cognitive processing associated with SVA. These attentional disturbances represent a core cognitive vulnerability, as attention underpins and shapes multiple higher-order cognitive functions (Yan et al., 2024).

Higher-order cognitive functions, such as memory, are also affected in individuals with SVA. Studies report reductions in working memory capacity, diminished verbal recall, and disruptions in continuous memory encoding, particularly in tasks that require sustained and controlled cognitive engagement (Al-Leimon et al., 2025; Li et al., 2025; Rahayu et al., 2025). Beyond memory, research has increasingly examined how SVA impacts decision-making. Across several studies, SVA is associated with altered reward-based judgements, reflected in increased risk-taking tendencies, heightened sensitivity to immediate rewards (Zhang & Li, 2025), and a documented reduction in loss aversion (Liu et al., 2025a). These findings indicate that SVA may recalibrate reward-processing systems in ways that prioritise immediate gratification. Yet, this influence is not uniform across all decision-making domains;

for example, ambiguous decision-making remains largely unaffected (Ye et al., 2025). Such distinctions suggest that SVA primarily affects motivationally charged evaluative processes rather than global cognitive judgement.

The changes in decision-making associated with SVA are closely tied to self-control. Individuals with lower baseline self-control are less able to regulate reward-driven impulses, making them more susceptible to habitual, automatic, and compulsive short-video viewing (Liu et al., 2025b; Wu et al., 2024; Yan et al., 2024; Zhu & Fong, 2025). At the same time, protective cognitive factors have been identified: mindfulness practices strengthen executive functioning and reduce SVA symptoms (Li et al., 2024; Shuja et al., 2025). These findings collectively highlight the reciprocal relationship between self-regulatory functioning and the compulsive, reward-oriented nature of SVA engagement.

Beyond core cognitive systems, emerging evidence suggests that SVA may also affect creative functioning. Excessive short-video consumption has been associated with reduced creative self-efficacy, potentially limiting individuals' confidence in their ability to generate novel ideas or engage in complex problem solving (Lin et al., 2023). Although further research is required, these findings point to cognitive consequences that extend beyond attentional and self-regulatory domains.

These converging cognitive impairments provide a critical foundation for understanding SVA's academic consequences. Across multiple studies, higher SVA levels have been consistently associated with lower academic performance, reduced study time, and poorer GPA (Al-Leimon et al., 2025; Alfatih et al., 2024; Guo & Chai, 2024; Xie et al., 2023; Xu et al., 2023). While individual studies attribute academic difficulties to different mechanisms-including attentional fragmentation, motivational depletion, and weakened executive control-the literature broadly agrees that the academic toll of SVA reflects the cumulative impact of its multi-layered cognitive disruptions. Thus, academic underperformance emerges not as an isolated outcome but as a downstream manifestation of interconnected deficits across attention, memory, decision-making, and self-regulatory systems.

Despite extensive research examining the impact of problematic short-video use on the cognitive functioning of adolescents and young adults, the existing evidence remains fragmented, with considerable variation in sample characteristics, cognitive domains assessed, and methodological rigour. Although the literature consistently links short video addiction (SVA) to cognitive deficits, previous systematic reviews have predominantly focused on mental health outcomes (Conte et al., 2025; Galanis et al., 2025; Jain et al., 2025; Paulus et al., 2023), often overlooking a structured synthesis of cognitive implications. Given the increasing prevalence of short-video use among youth and the potential for long-term cognitive consequences, a systematic review is needed to consolidate current evidence, identify knowledge gaps, and inform future research and intervention strategies. Accordingly, the present review aims to systematically synthesise empirical evidence on the influence of SVA on cognitive functions among adolescents and young adults, with a focus on identifying and summarising the cognitive domains examined, assessing the methodological quality, identify gaps and suggest directions for future research.

Methods

This review adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020.

Eligibility criteria

To be included in the review, articles must have assessed problematic or addictive short video use among adolescents or young adults aged 10 to 35. In addition, the studies should have assessed at least one of the foremost cognitive outcomes, including executive function, cognitive control, attention, working memory, or decision-making. The articles should have been published in peer-reviewed journals and in the English language. This review excluded non-empirical studies, including literature reviews, commentaries, editorials, and theoretical papers, as well as grey literature, conference abstracts, and non-peer-reviewed articles.

Table 1. Search terms used in the search string for the databases.

Variable	Search terms
Short-form video	Short-form video OR Short video OR TikTok OR YouTube shorts OR Reels
Addiction	Addiction OR problematic use
Cognitive outcome	Attention OR Executive function OR Cognitive control OR Decision-making OR Memory

Information sources and search strategy

Major electronic databases PubMed, PsycINFO, ScienceDirect, Scopus, and EBSCOhost were searched using relevant keywords to identify articles for the review. Table 1 displays the search terms that were implemented.

The search was completed by the end of May 2025, using a systematically developed search strategy. Minor modifications were made to the search string based on the requirements of the databases. Reference lists of the articles obtained from database screening were manually reviewed to identify any additional relevant studies.

Selection of studies

All articles from the database search were imported into Rayyan, a platform for article screening, and duplicates were eliminated. The remaining studies were evaluated in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 standards (Page et al., 2021).

Following the elimination of duplicates, MS, AS, and PY evaluated the titles and abstracts of the remaining publications to identify and choose studies that fulfilled the inclusion and exclusion criteria. This was confirmed by RP and VR. A subsequent secondary screening was conducted by AS and PY, and validated by MS and VR. The complete text of the article was evaluated according to the eligibility criteria. Exclusion reasons were documented at each stage.

Data extraction

Upon finalising the articles, data extraction was conducted utilising a structured template on Google Sheets. DV and TR extracted pertinent information, including study characteristics (first author's name, year of publication, study location, study design, sample size), participant characteristics (age and gender), cognitive outcomes assessed, and major findings, which were subsequently verified by RP. The disagreements and doubts about studies were cleared by discussion between the particular reviewer and verifier.

Quality assessment and data synthesis

VR and RP independently evaluated the quality of the studies utilising the JBI critical appraisal tool designed for analytical cross-sectional and quasi-experimental studies, as published by the Joanna Briggs Institute in 2023. Each item received a score of one for a 'Yes' response, while 'No' and 'Unclear' responses were assigned a score of zero. Disagreements between the two reviewers regarding the qualitative evaluation of studies were resolved by obtaining validation from MS and AS. Scores below five were classified as low, those ranging from five to seven as moderate, and scores exceeding seven as high. A synthesis of the narrative and thematic data extracted was conducted to present the findings of the review.

Results

A total of 959 articles were retrieved from the databases. Sixty-two duplicate articles were eliminated, leaving 897 articles for title and abstract screening. In the title and abstract screening phase, 855 articles were excluded for failing to meet the study's criteria, while 42 articles were included for full-text screening. In the full text screening, 24 articles were excluded, while 18 articles meeting all eligibility criteria were included in the review. Furthermore, 12 articles were acquired through manual search. Of these, seven articles were deemed ineligible for the reasons outlined in Figure 1, while the remaining five eligible articles

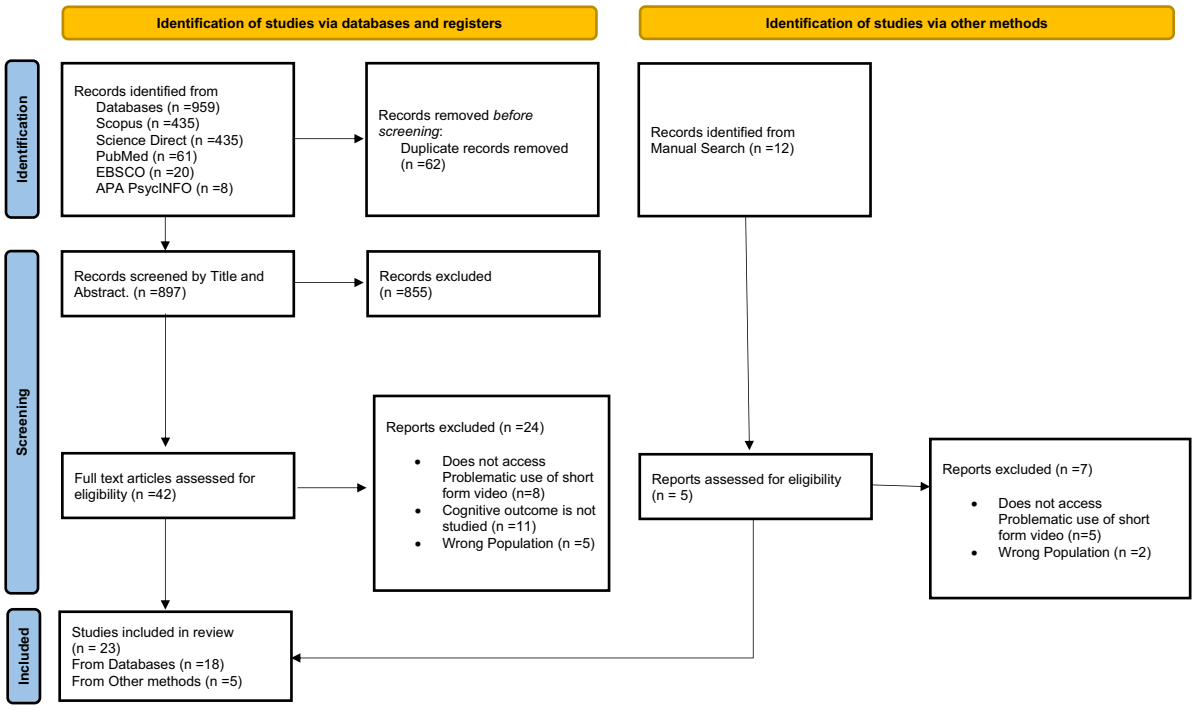

Figure 1. Prisma 2020 flow diagram for selection of studies. Source: Page MJ, et al. BMJ 2021;372:n71. doi: 10.1136/bmj.n71. This work is licensed under CC BY 4.0. To view a copy of this license, visit https://creativecommons.org/licenses/by/4.0/.

were incorporated into the review. A total of 23 articles from the database and manual search were included in the final review. Figure 1 presents the PRISMA flow diagram illustrating the processes of identification, screening, and selection of articles for this review.

Characteristics of the reviewed studies

Table 2 (Supplementary material) summarises the reviewed papers. A total of 21 studies utilised cross-sectional and correlational designs, whereas two studies employed a quasi-experimental design (Chen et al., 2022; Jiang et al., 2024). The total sample size across all studies was 22,122, with individual samples ranging from 36 to 4,750 adolescents or young adults. The age of participants varied from 11 to 31 years. Twenty studies were conducted in China, two in India (Pranathi & Jacob, 2025; Singh & Prusty, 2025), and one in Jordan (Al-Leimon et al., 2025). While the majority of studies have concentrated on SVA broadly, four studies (Jianfeng et al., 2024; Qin et al., 2022; Sha & Dong, 2021; Su et al., 2021) specifically examined TikTok addiction.

Quality assessment

The risk of bias was assessed using the JBI critical appraisal tool (see supplementary material). Most cross-sectional studies ($n = 18$) were of moderate quality, two were high (Liu et al., 2025a; Pan et al., 2025), and one was low (Qin et al., 2022). Both quasi-experimental studies were high quality (Chen et al., 2022; Jiang et al., 2024). Only three studies reported confounding variables, while several lacked clear sampling methods or used convenience sampling, raising concerns about selection bias (See Tables 3 and 4 in supplementary material).

Cognitive outcomes assessed

The findings of the review revealed that attention and self-control are the cognitive functions most strongly affected by SVA. Other outcomes such as memory, negative cognitive bias, decision making, loss of aversion and cognitive engagement were reported less frequently.

Attention

The examined studies consistently demonstrated that short video addiction (SVA) has a substantial adverse effect on attentional functioning.

SVA is significantly associated with reduced attentional control, as demonstrated by studies conducted by Xie et al. (2023), Ye et al. (2025), Zhou et al. (2024), and Singh and Prusty (2025). The correlations were low to moderate in strength ($r = -0.23$, $p < .01$; $r = -0.56$, $p < .01$; $\beta = 0.43$, $p < .001$; $r = -0.209$, $p = .006$). These results suggest that individuals' capacity to effectively manage their attentional resources is compromised by their addictive engagement with short videos. Extending this evidence, Zhuo et al. (2024) reported a moderate negative correlation between SVA and attention focus ($r = -0.60$, $p < .01$) and a low negative correlation with attention shift ($r = -0.38$, $p < .01$). This suggests that SVA impedes the capacity to flexibly reorient attention when necessary and the ability to sustain attention on a single task. Similarly, Pranathi and Jacob (2025) demonstrated that SVA is substantially correlated with a reduced attention span ($r = -0.295$, $p < .001$).

These results were also supported by four experimental investigations. Chao et al. (2023) reported that SVA significantly mediated enjoyment and concentration ($p < .01$), with addicted users exhibiting pronounced attentional problems ($r = .416$, $p < .01$). Qin et al. (2022) found that exposure to TikTok videos disrupted concentration in flow experiences. Chen et al. (2022) similarly observed that individuals with higher SVA exhibited increased distraction and attentional deficiencies during and after video viewing. Wu et al. (2025) emphasised that attentional bias acted as a mediator in the relationship between SVA and adolescent depression, in addition to acute attentional disruptions ($\beta = 0.084$, 95% CI). Their research demonstrated that depressive symptoms were exacerbated by less attention to positive cues and increased attention to negative cues, thereby establishing a connection between attentional impairments related to SVA and mental health outcomes.

In general, the accumulated evidence indicates that SVA significantly undermines a variety of attentional functions, such as attentional control, focus, shift, span, concentration, and bias.

Self-control

The results of the studies in this review consistently demonstrated that self-control functions as a significant protective factor against short video addiction (SVA). Throughout all seven studies, self-control was identified as a negative predictor of addictive use, with correlations ranging from low to strong intensity. Additionally, numerous studies have confirmed its mediating function between psychosocial factors and SVA. The study conducted by Zhang et al. (2024) investigated the influence of social exclusion and boredom on addictive tendencies. The results indicated that increased levels of boredom predicted a greater level of SVA ($p < .001$), while higher levels of self-control substantially reduced these effects ($p < .01$). Liu et al. (2025b) demonstrated that individuals with a stronger future time focus were more likely to initiate self-control ($p < .01$), which subsequently decreased the likelihood of SVA ($p < .01$). This finding underscores the forward-looking and regulatory role of self-control in addiction prevention.

Further, self-control has been identified as a critical mechanism that connects personality and behavioural variables with decreased addiction. Jianfeng et al. (2024) discovered that physical activity indirectly reduced SVA by improving self-control ($r = -0.48$, $p < .01$). Similarly, Wu et al. (2024) found that self-control mediated nearly half (44.65%) of the association between proactive personality and lower SVA ($r = -0.453$, $p < .01$). This suggests that personality traits that are conducive to regulation are effective largely because they bolster self-control. These findings were also corroborated by neurocognitive evidence: Su et al. (2021) found that problematic TikTok use was inversely correlated with reduced self-control ($r = -0.279$, $p < .001$) and was accompanied by heightened reward sensitivity in neural pathways. This suggests a biological vulnerability in low self-control users.

Self-control's protective function was additionally validated in high-risk adolescent populations. Liu et al. (2022) discovered that self-control was a substantial negative correlate of SVA, even among adolescents who experienced parental neglect and low school connectedness ($r = -0.38$, $p < .001$).Similarly, Jiang et al. (2024) reported that addicted users scored considerably lower on self-control measures than their non-addicted peers ($t = -2.80$, $p = .007$, $d = -0.73$), thereby confirming a substantial self-control deficit that is associated with increased impulsivity and addictive behaviour.

Overall, SVA had a significant inverse relationship with self-control, as measured by dimensions such as inhibitory control, proactive regulation, and impulse control. Those with greater self-control were less likely to engage in persistent, excessive video use, whereas those with lower self-control were more susceptible to compulsive overuse, impulsivity, and distraction.

Other cognitive outcomes

The review of seven studies among Jordanian and Chinese university students underscored the more extensive cognitive repercussions of short video addiction (SVA), which surpass attention and self-control. Evidence indicated that SVA influenced a variety of cognitive vulnerabilities, including decision-making, learning engagement, loss aversion, cognitive bias, memory, and self-evaluation. Using fNIRS, Zhang and Li (2025) examined the influence of SVA on risky decision-making. They found that students with higher SVA exhibited more robust activation in the orbitofrontal cortex and the frontopolar area when exposed to short video stimuli. The results suggested that addictive short video use disrupts decision-making by increasing reward sensitivity and cognitive processing toward immediate gratification.

In the educational domain, Ye et al. (2023) found that the misuse of short-video platforms like TikTok had a detrimental impact on behavioural engagement ($\beta = -0.50$, $t = -12.94$), which in turn affected affective engagement ($\beta = 0.75$, $t = 23.47$) and cognitive engagement ($\beta = 0.65$, $t = 21.17$). Similarly, Liu et al. (2025a) investigated loss aversion using inter-subject representational similarity analysis and Drift Diffusion Models. They discovered a significant negative correlation between SVA symptoms and loss aversion ($r = -0.378$, $p = .023$). In addition, neuroimaging findings demonstrated that SVA modifies neural mechanisms of risk and reward evaluation, as evidenced by altered brain activation during profit and loss processing.

Other research has demonstrated cognitive distortions associated with SVA. Negative cognitive bias was found to predict higher levels of SVA ($b = 0.153$, 95% CI [0.026, 0.281], $p < .01$) by Yue et al. (2024). The relationship was mediated by loneliness and inadequate social support. This shows that maladaptive cognitive patterns not only predispose individuals to SVA but are also reinforced by it, thereby establishing a cycle of reliance on short video platforms. Cognitive impairments in memory were also documented. In 2021, Sha and Dong discovered that TikTok use disorder was linked to working memory deficits ($p < .001$), as well as depression, anxiety, and stress. Al-Leimon et al. (2025) found that high levels of SVA predicted poor memory performance ($r = -0.31$, $p < .01$), with attentional deficits serving as a partial mediator (indirect effect, $p < .05$). These findings emphasise the long-term effects of SVA on cognitive efficiency, demonstrating that the impairment extends beyond temporary distraction and may influence core memory processes.

In conclusion, research has investigated the potential for self-evaluation to result from SVA. A chain mediation model was employed by Pan et al. (2025) to evaluate the potential correlation between adolescents' short video consumption and central self-evaluation via maladaptive cognitions and information overload. The direct effect was not statistically significant; however, significant indirect effects were observed through information saturation (0.077), maladaptive cognitions (0.208), and a sequential pathway that connected both (0.025). This implies that the excessive use of short video platforms indirectly diminishes self-evaluation by inundating users with information and reinforcing maladaptive online thought patterns.

Combined, the systematic review demonstrates that SVA has a substantial impact on a broad range of cognitive functions, in addition to its well-established effects on attention and self-control. From diminished self-evaluation and heightened cognitive biases to impairments in decision-making, learning engagement, and memory, SVA poses a multifaceted threat to cognitive functioning in adolescents and young adults.

Discussion

Though there are multiple systematic reviews carried out connecting SVA and mental health and well-being outcomes, systematic reviews are sparse when it comes to SVA and cognitive outcomes. The present review of the 23 studies reveals that attention and self-control are the most impacted cognitive functions due to SVA. The less frequent cognitive outcomes identified in the review include memory, negative cognitive bias, decision-making, loss of aversion and cognitive engagement.

The review shows a low to moderate link between attention control and SVA, with greater distraction and attention problems during and after viewing (Chen et al., 2022; Singh & Prusty, 2025; Xie et al., 2023; Zhou et al., 2024), which threatens overall attentional capacity. These findings support previous literature from EEG studies showing a significant negative correlation between short-video addiction and executive control, indicating reduced attention-related activity in the prefrontal cortex (Yan et al., 2024). They also align with Walla and Zheng (2024) observation that heavy short-video users exhibit lower event-related potentials, a brain marker of attention, suggesting impaired ability to shift focus. This pattern is further supported by neural evidence from Su et al. (2023), who showed that watching preferred short videos activates the amygdala while deactivating prefrontal control regions, with individuals low in trait self-control exhibiting stronger suppression of these executive systems. Such bottom-up limbic dominance helps explain why attention is captured so effortlessly by short videos and why users struggle to disengage.

The findings of the present review also noted that watching TikTok videos significantly mediates enjoyment and concentration and TikTok addiction is strongly linked to attention problems and other mental health issues (Chao et al., 2023; Qin et al., 2022), attentional bias also mediates the relationship between short-video addiction and depression (Wu et al., 2025), echoing Xiao et al. (2022) results showing that attention bias mediates the link between social media addiction and depression in adolescents. However, the literature presents mixed results. Active usage of short videos, such as creating or interacting with content, enhances life satisfaction and positive affect by alleviating social anxiety, whereas passive usage, such as merely scrolling, reduces life satisfaction and positive affect and increases negative affect (Zhai et al., 2024). Short videos can also positively influence the mental health of elderly individuals by improving intergenerational relationships and leisure habits (Zhang et al., 2024). The impact on subjective well-being varies across users and ages, and these mixed results could be due to multiple factors including the type, length, and content of the video, intelligent push functions, page design, frequency of use, and users' professions or demographics (Ying & Phu-ngamdee, 2023).

The review highlights that there is a negative association between SVA and attention span (Pranathi & Jacob, 2025) and that short video use indirectly harms adolescents' core self-evaluation by causing information overload and maladaptive internet cognition (Pan et al., 2025). Similarly, Ye et al. (2023) found that excessive short-video use undermines students' attention and participation, reduces emotional and cognitive engagement, and leaves them feeling their learning was ineffective. These findings align with previous research indicating that high short-video consumption can negatively impact academic achievement due to reduced attention span (Asif & Kazi, 2024). This also supports earlier research showing that short video use leads to information overload, loss of time awareness, attention deficits, and increased academic procrastination (Caponnetto et al., 2025; Dang, 2024). Furthermore, the findings are in line with Mahmoudi et al. (2024), who found that watching short videos reduces pupil diameter, an indicator of decreased attention and cognitive processing, which when occurring before study sessions can impair learning. Supporting this, evidence from problematic smartphone use reveals impaired attentional control, hyperactivation of frontoparietal regions, and weakened ventral attention network connectivity, indicating that individuals with high digital dependence recruit attention inefficiently and struggle to filter irrelevant stimuli (Choi et al., 2021). Yet, research also shows that when used intentionally in structured learning environments, short videos can enhance cognitive engagement, support attentional focus, and improve academic outcomes, suggesting that the medium's effects depend on context, regulation, and instructional design (Wang, 2025b).

The review also revealed that short-video addiction is linked to impairments in working memory (Sha & Dong, 2021) and attention partially mediates the negative effect of short-video addiction on memory (Al-Leimon et al., 2025). This echoes findings that excessive short video usage and screen time can impair working memory functions (Liu, 2022; Shamloul, 2024) and aligns with literature showing that video game addiction, is correlated with poorer memory, reduced attention, and diminished cognitive and academic abilities in children (Farchakh et al., 2020). However, some findings also indicate that exposure to short videos can enhance certain cognitive skills in children such as creativity, curiosity, and vocabulary acquisition, highlighting a dual effect where structured, age-appropriate content may support development while unregulated or harmful content disrupts cognitive and brain health (Putri et al., 2024).

Findings on decision-making and risk-taking behaviour suggest that students addicted to short videos are more likely to make risky decisions and act impulsively (Zhang & Li, 2025) also another study found

negative correlation between addiction severity and loss aversion, indicating that addicted students were less sensitive to losses and were more inclined to take risks (Liuet al., 2025a). This aligns with previous findings showing that smartphone addicts exhibit impaired decision-making under ambiguity, similar to deficits observed in chemical and behavioural addictions such as alcohol dependence, gambling disorders, and compulsive buying (Khoury et al., 2019), consistent with patterns seen in SVA. In contrast, earlier work on Internet addiction reported better decision-making and reduced risk-taking, likely due to intact implicit emotional learning (Ko et al., 2010). While these contrasting findings suggest variability across types of digital addiction, the current results are supported by neurological evidence showing that exposure to short-video cues heightens activation in reward-related regions, particularly the right orbitofrontal cortex (OFC) and right frontopolar area (FPA). This suggests increased reward sensitivity that may drive more impulsive and risky behaviour, helping to explain why short-video addiction can impair decision-making (Zhang & Li, 2025).

Several articles in the review highlighted factors that contribute to SVA. Yue et al. (2024) found that individuals with higher negative cognitive bias are more prone to SVA. Increased boredom projected greater addiction, while stronger self-control significantly reduced it (Zhang et al., 2024); moreover, students focused on the past or present are more likely to develop SVA, whereas a future-oriented mindset reduces this risk, with initiating self-control mediating the positive effect of future time focus by lowering addiction risk (Liu et al., 2025b). Those with lower self-control are vulnerable to SVA and higher impulsivity (Jiang et al., 2024; Su et al., 2021), which aligns with Błachnio et al. (2023), who found that self-control negatively relates to problematic Internet use, smartphone use, and Facebook use. Neuroscientific findings reinforce this, showing that preference-based short-video viewing dampens prefrontal regions responsible for rule maintenance and self-awareness, thereby weakening top-down self-control mechanisms (Su et al., 2023).

Furthermore, the review highlighted the role of various factors, particularly enhanced self-control, in reducing SVA. For example, physical exercise reduces adolescent SVA partly by boosting self-control, with cognitive emotion regulation strengthening these effects (Jianfeng et al., 2024). This aligns with previous research showing that physical exercise reduces adolescent cell phone dependence through improved self-control (Zhang et al., 2022) and supports findings that both self-control and physical activity have long-term protective effects against problematic smartphone use (Zhao et al., 2024). The review found that proactive personality traits lower addiction risk, with trait self-control identified as a significant protective factor (Liu et al., 2022; Wu et al., 2024). This aligns with previous research identifying key self-control strategies for managing social networking use and showing that the difficulty in applying these strategies partly explains how trait self-control influences social networking addiction severity through habitual use (Brevers & Turel, 2019).

The review shows that short-video addiction mainly disrupts attention and self-control, with additional effects on memory, decision-making, and cognitive engagement. Neurological evidence suggests reduced prefrontal activity and increased limbic response, explaining why users are easily distracted and struggle to disengage. The effects of short videos vary with usage patterns, content, and individual traits such as self-control and impulsivity, and strategies like physical exercise and cognitive regulation can help reduce harm. Overall, SVA presents clear cognitive risks, but structured use and personal regulation can mitigate these effects.

Implications, limitations and future prospects

Among the 23 articles reviewed, 20 studies were conducted in China, two in India, and one in Jordan. Literature shows SVA is a global issue, with TikTok projected to have 955.3 million users worldwide in 2025 (Statista, 2025) and YouTube Shorts attracting over two billion monthly users and 70 billion daily views (Kumar, 2025). Studies are scarce in other countries, though only English language studies are included in the current review. This implies the need to replicate similar studies in other countries to understand different usage patterns, the role in the development and severity of SVA, and also to understand the relationship between cognitive outcomes and SVA in diverse populations, which may in turn help promote healthier use of these platforms. Future studies should consider cultural, social, and demographic factors, as patterns of short video addiction and its cognitive impact may vary across countries, age groups, genders, and socioeconomic backgrounds.

The majority of the studies focused on SVA's effect on attention as the primary cognitive outcome. It is suggested that further research can be undertaken to understand SVA's role in affecting other cognitive attributes such as memory impairment, decision-making, loss aversion and cognitive engagement which have limited number of studies. SVA effect on other cognitive outcomes such as language, problem-solving, critical thinking, and reasoning can be key areas of focus in the future. Also, future research can also delve into studying whether the genre and duration of short videos has an effect on cognitive outcomes.

The review also shows that the majority of the studies are cross-sectional and correlational, apart from two quasi-experimental studies. As most studies are correlational, the evidence does not directly establish causation, which means that there could be multiple factors contributing to the cognitive outcomes, and not just SVA. This highlights the need for conducting more true experiments and longitudinal studies to establish cause-effect relationships. Furthermore, most of the cross-sectional studies included in the review were of moderate quality, with a few rated as high and one as low. The two quasi-experimental studies were both high quality, which further implies the need for conducting more experimental research.

The present review used five major databases including PubMed, PsycINFO, ScienceDirect, Scopus, and EBSCO along with manual searches to extract studies. Despite rigorous adherence to PRISMA guidelines and efforts to include all relevant work, some critical studies from other databases may have been missed, which could have contributed to the findings and remains a limitation. Additional limitations include language bias, as only English-language studies were included, and publication bias, since unpublished or negative-result studies may have been overlooked. Variations in measurement tools for SVA and cognitive outcomes may also limit comparability, while reliance on self-reported video usage can introduce reporting inaccuracies.

In the review, it was found that a number of studies did not report information on sampling methods, and others relied on convenience sampling, which could increase the risk of selection bias. This calls for future scholars to conduct more research using random sampling methods to enhance the generalisability of the findings to a larger population. Future research could also examine the role of video genre, duration, and context of use, as well as interventions promoting digital literacy and self-regulation.

Conclusions

The present review revealed the impact of SVA or problematic short-video usage on cognitive outcomes, factors contributing to SVA, and protective factors that may help reduce addiction. It was found that SVA negatively impacts most cognitive outcomes such as attention and related variables, impairs memory, decision-making, loss aversion and cognitive engagement. This highlights the problematic results of such addiction, which may need to be addressed by psychologists, media practitioners, policymakers and scholars who can create ideal intervention strategies to help both adolescents and young adults manage their short video consumption habits. The review also found that self-control shares a negative relationship with SVA, suggesting that increasing self-control may help reduce addiction. This points to the need for more intervention-based research to explore this further and use it effectively for reducing SVA. It was also found that physical exercise reduces adolescent SVA partly by boosting self-control. Furthermore, the review identified certain factors contributing to SVA, such as higher negative cognitive bias, increased boredom, and a focus on the past or present. It is pertinent to tackle these factors in order to reduce SVA among the adolescent and young adult demographic, thereby, helping them gain better traction in cognition.

Author contributions

AM—Conception and design of the review, development of the review protocol, study selection, data extraction, data synthesis and interpretation, draughting the manuscript, and critical revision for important intellectual content. VR—Conception and design of the review, study selection, data extraction, data synthesis and interpretation, draughting the manuscript, and critical revision for important intellectual content. RP—Conception and design of the review, study selection, data extraction, data synthesis and interpretation, draughting the manuscript, and critical revision for important intellectual content. PR—Study selection, data extraction, data synthesis and interpretation, draughting the manuscript. AH—Study selection, data extraction, data synthesis and interpretation, draughting the manuscript. DM - Study selection, data extraction, data synthesis and interpretation, draughting the manuscript. TM—Study selection, data extraction, data synthesis and interpretation, draughting the manuscript. All authors approved the final version and agreed to be accountable for all aspects of the work, including accuracy and integrity.

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Data availability statement

The data that support the findings of this study are available from corresponding author upon reasonable request.

Ethics statement

This review article is based on previously published studies and does not involve any new studies with human participants or animals conducted by the authors. Therefore, ethical approval and informed consent were not required.

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